

Name: <u>Solution</u>	ID:	Section:
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1.CO1	Divide $(2342)_7$ by $(14)_7$. Show all the working steps.	4
2.CO1	Add $(89)_{10}$ with $-(45)_{10}$ in 10 bits using 2's complement number system. Justify whether there is an overflow or not.	5
3.CO1	Suppose you are going on a picnic with $(23)_{16}$ other people. Each person, including yourself, needs to contribute $(320)_6$ taka for the picnic expenses. After the picnic trip was over, you found out that $(516)_9$ was spent on food and $(101001100)_2$ was spent on transportation. How much money was left after paying all the expenses? Show your answer in decimal.	6

$$\begin{array}{r}
 141 \\
 14 \overline{) 12342} \\
 \underline{-14} \\
 064 \\
 \underline{-62} \\
 01229 \\
 \underline{-14} \\
 5
 \end{array}$$

$$\begin{aligned}
 14 \times 1 &= 14 \\
 14 \times 2 &= 31 \\
 14 \times 3 &= 45 \\
 14 \times 4 &= 62
 \end{aligned}$$

$$\begin{aligned}
 \text{Quotient} &= (141)_7 \\
 \text{Remainder} &= (5)_7
 \end{aligned}$$

$$\begin{array}{r}
 2 \overline{) 89} \\
 2 \overline{) 44} 1 \\
 2 \overline{) 22} 0 \\
 2 \overline{) 11} 0 \\
 2 \overline{) 5} 1 \\
 2 \overline{) 2} 1 \\
 2 \overline{) 1} 0 \\
 \hline
 0 1
 \end{array}$$

$$\begin{array}{r}
 2 \overline{) 45} \\
 2 \overline{) 22} 1 \\
 2 \overline{) 11} 0 \\
 2 \overline{) 5} 1 \\
 2 \overline{) 2} 1 \\
 2 \overline{) 1} 0 \\
 \hline
 0 1
 \end{array}$$

$$(45)_{10} = (101101)_2$$

$$(89)_{10} = (1011001)_2$$

$$(89)_{10} \text{ in 10-bits} = (0001011001)_{25}$$

$$(+45)_{10} = (0000101101)_{25}$$

$$\begin{array}{r}
 (-45)_{10} = 111010010 \\
 \phantom{(-45)_{10} = } + 1 \\
 \hline
 = 111010011
 \end{array}$$

$$\begin{array}{r}
 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \\
 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \\
 + \ 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1 \\
 \hline
 \textcircled{X} \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0
 \end{array}$$

$$\text{Result} = (0000101100)_{25}$$

No overflow since we are adding two different signed numbers.

$$\begin{array}{l}
 3) \quad \boxed{
 \begin{array}{l}
 (1F)_{16} = 1 \times 16^1 + 15 \times 16^0 \\
 = (31)_{10} \\
 \text{Total } P
 \end{array}
 }
 \end{array}$$

$$\begin{aligned}
 (23)_{16} &= 2 \times 16^1 + 3 \times 16^0 \\
 &= (35)_{10}
 \end{aligned}$$

$$\text{Total people} = 35 + 1 = 36$$

$$\begin{aligned}
 \text{Each person contribution} &= (320)_6 \\
 &= 3 \times 6^2 + 2 \times 6^1 + 0 \times 6^0 \\
 &= (120)_{10}
 \end{aligned}$$

$$\text{Total money} = 36 \times 120 = (4320)_{10}$$

$$\begin{aligned}
 \text{Food} &= (516)_9 = 5 \times 9^2 + 1 \times 9^1 + 6 \times 9^0 \\
 &= (420)_{10}
 \end{aligned}$$

$$\begin{aligned}
 \text{Transport} &= (101001100)_2 \\
 &= 1 \times 2^8 + 1 \times 2^6 + 1 \times 2^3 + 1 \times 2^2 \\
 &= (332)_{10}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total money left} &= 4320 - 420 - 332 \\
 &= (3568)_{10}
 \end{aligned}$$